

The Nernst Equation and Cell Potential

Answer key with worked solutions

A. Ideas

1. Battery voltage.

As the cell runs, reactants are used up and products build up, so the reaction quotient **Q increases**. By the Nernst equation, $E = E^\circ - \frac{0.0592 \text{ V}}{n} \log Q$, so **E decreases**. When the reaction reaches equilibrium, $Q = K$ and **E = 0**: the battery is flat.

B. Using the Nernst equation

2. Zn-Cu cell.

$$\begin{aligned} E^\circ &= 1.10 \text{ V}, n = 2, Q = \frac{0.10}{2.0} = 0.050 \\ E &= 1.10 \text{ V} - \frac{0.0592 \text{ V}}{2} \log 0.050 \\ &= 1.10 \text{ V} + 0.0385 \text{ V} = \mathbf{1.14 \text{ V}} \end{aligned}$$

Larger than E° : Q is less than 1 (more reactant Cu^{2+} than product Zn^{2+}), so the reaction has a greater driving force.

3. Ag-Cu cell.

$$\begin{aligned} \text{(a)} \quad E^\circ &= 0.80 \text{ V} - 0.34 \text{ V} = \mathbf{0.46 \text{ V}}; n = \mathbf{2} \text{ (2 Ag}^+ \text{ each gain 1 e}^- \text{)} \\ \text{(b)} \quad Q &= \frac{[\text{Cu}^{2+}]}{[\text{Ag}^+]^2} = \frac{1.0}{(0.010)^2} = 1.0 \times 10^4 \\ E &= 0.46 \text{ V} - \frac{0.0592 \text{ V}}{2} \log(1.0 \times 10^4) \\ &= 0.46 \text{ V} - 0.118 \text{ V} = \mathbf{0.34 \text{ V}} \end{aligned}$$

4. Silver concentration cell.

(a) The **dilute** side (0.0010 M) is the anode: Ag is oxidized there, raising its Ag^+ concentration.

$$\text{(b)} \quad E^\circ = 0, n = 1, Q = \frac{0.0010}{0.10} = 0.010; E = 0 - 0.0592 \text{ V} \times \log 0.010 = \mathbf{+0.118 \text{ V}}$$

C. E° , ΔG° and K

5. ΔG° and K .

$$(a) \Delta G^\circ = -2 \times 96\,485 \text{ C/mol} \times 0.46 \text{ V} = -8.88 \times 10^4 \text{ J/mol} = \mathbf{-88.8 \text{ kJ/mol}}$$

$$(b) \log K = \frac{2 \times 0.46 \text{ V}}{0.0592 \text{ V}} = 15.54, \text{ so } K = \mathbf{3.5 \times 10^{15}}$$

6. E° from K .

$$E^\circ = \frac{0.0592 \text{ V}}{1} \log(1.0 \times 10^5) = 0.0592 \text{ V} \times 5.0 = \mathbf{0.30 \text{ V}}$$

7. Negative E° .

$\Delta G^\circ = -nFE^\circ$ is **positive**, and K is **less than 1**: the reaction is not spontaneous under standard conditions, and reactants are favoured at equilibrium.

D. Challenge

8. When $E = 1.00 \text{ V}$.

$$1.00 \text{ V} = 1.10 \text{ V} - \frac{0.0592 \text{ V}}{2} \log Q$$

$$\log Q = \frac{2 \times 0.10 \text{ V}}{0.0592 \text{ V}} = 3.38$$

$$Q = \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]} = \mathbf{2.4 \times 10^3}$$

The voltage falls only slowly at first: zinc ions must outnumber copper ions by about 2400 to 1 before E drops by 0.10 V.